TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Practical Processing of Different Counts of Cotton and Jute Yarns and Analysis of Lap, Sliver and Roving Properties

One combined study sheet: processing trials for two or three target counts on the cotton line, a parallel trial on the jute line, and laboratory analysis of the intermediates — lap, sliver and roving.

YM-4102 · PROCESS CONTROL IN SPINNING SESSIONAL · COMBINED STUDY SHEET · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: (i) to process cotton through the full spinning line — blowroom, card, draw frame, simplex (speed frame) and ring frame — set to produce yarns of two or three different target counts; (ii) to process jute through its own line — batching and softening, breaker card, finisher card, drawing frames and the jute spinning frame; and (iii) to analyse the intermediates at each stage — lap weight per metre and lap variation, sliver hank and evenness, roving hank and twist — so that a fault in the final yarn can be traced back to the stage where it was created.

The governing idea is **process control**: yarn count and quality are not made at the ring frame alone. Each machine hands its product to the next, and each intermediate has a target value — a hank, a weight per unit length, a variation limit — to be checked before the material goes forward. A heavy lap makes a heavy card sliver; a heavy sliver survives doubling only in part; and an off-hank roving can be corrected at the ring frame only within its change-wheel range. Measuring the intermediates is therefore not an extra exercise: it is the processing, done properly.

1.1 The quantities that are controlled

- **Hank** — the indirect count of a sliver or roving: the number of 840-yard lengths in one pound. Hank = (length in yards ÷ weight in pounds) ÷ 840. A finer intermediate has the higher hank number.
- **Yarn count, Ne** — the same idea applied to yarn: the number of 840-yard hanks in one pound.

$$\text{Ne} = (\text{length in yd} \times 453.6) \div (840 \times \text{weight in g})$$

- **Draft** — the ratio by which a machine attenuates its feed: draft = delivered length ÷ feed length = feed weight per unit length ÷ delivered weight per unit length. Hank delivered = hank fed × draft, so every count change is, in the end, a draft change.
- **Coefficient of variation, CV%** — the standard deviation of a set of readings as a percentage of their mean; the standard single number for the evenness of a lap or sliver:

$$\text{CV\%} = (\text{standard deviation} \div \text{mean}) \times 100, \quad \text{with SD} = \sqrt{[\ \Sigma(x - \text{mean})^2 \div n\ ]}$$

- **Twist** — turns per unit length inserted in roving (light, only enough to let the roving wind, unwind and draft without stretching) and in yarn (much higher, and the main source of yarn strength).

1.2 Why the cotton and jute lines differ

Cotton is a short, fine, single-cell staple fibre supplied in bales, already reasonably clean. Its line therefore concentrates on opening, cleaning, carding and progressive drafting. Jute arrives as long reed bundles of bast fibre, stiff, lignified and held together by gummy matter. It must first be **softened** — treated with an oil-and-water emulsion and given time to condition (batching or piling) — and its cards must **break down** the long reeds into shorter fibre bundles rather than individualise fine single fibres. Jute is also counted differently: in pounds per spyndle, where one spyndle is 14,400 yards — a direct (weight-per-fixed-length) system, the opposite way round from the cotton hank system. The control principles are identical; the machinery and the arithmetic are not.

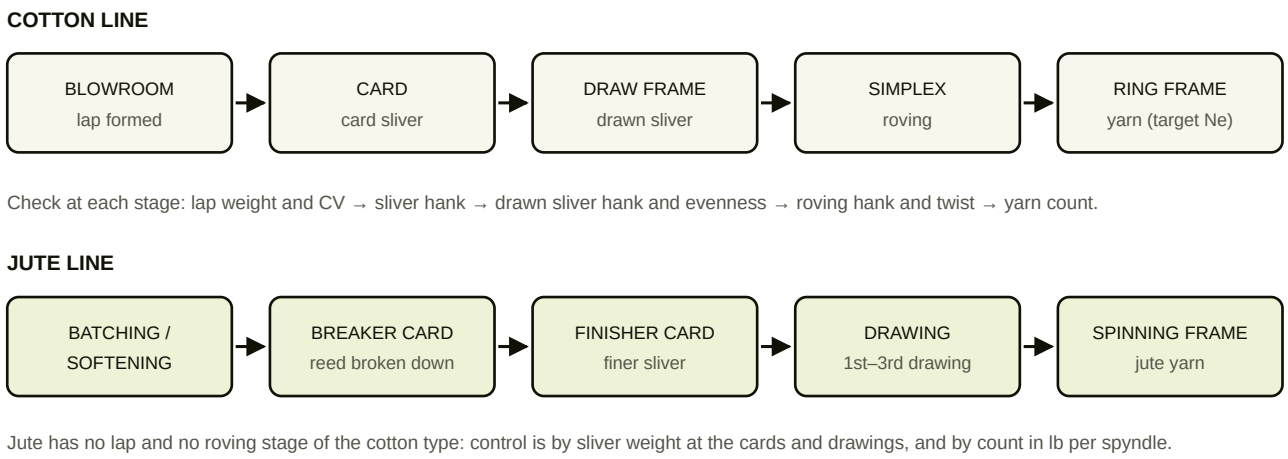


Figure 1 — The two processing lines side by side: cotton is drafted down through lap, sliver and roving; jute is softened, broken down and drawn.

2. Apparatus and procedure

2.1 Apparatus and instruments

ITEM	USE IN THIS SHEET

Cotton spinning line	Blowroom line with lap former, card, draw frame, simplex and ring frame, with the change wheels needed to alter draft and twist for each target count.
Jute processing line	Softening/batching arrangement with emulsion, breaker card, finisher card, jute drawing frames and jute spinning frame.
Wrap reel	Winds an exact measured length — a 120-yard skein (lea) of yarn, or a measured length of sliver or roving — for weighing. The reel removes the largest error in count work, the length error.
Analytical / precision balance	Weighs skeins, sliver and roving lengths and cut lap specimens to 0.01 g or better.
Evenness tester (concept)	Draws sliver, roving or yarn through a sensing head and records mass variation continuously, giving CV% or U% — used here in principle to judge sliver evenness beyond cut-and-weigh readings.
Lea strength tester (optional)	Breaks the whole 120-yard skein at once; lea strength combined with count gives the count–strength product, a quick yarn quality index.
Twist tester; lap scale and rule	The twist tester untwists a measured length of roving or yarn to count its turns; the lap scale and rule serve for cutting and weighing one-metre lap specimens.

2.2 Procedure — cotton line trials

1. **Plan the counts.** Fix two or three target yarn counts (a coarse, a medium and a finer count). Work backwards: for each, note the roving hank, the sliver hanks and the drafts to be set at each machine.
2. **Condition and open.** Run the blowroom on the selected mixing; check the lap as it is formed — weigh one-metre specimens and calculate lap weight per metre and its CV% (Section 3c) before the lap goes to the card. An out-of-limit lap is rejected here, not carried forward.
3. **Card.** Feed the lap, set the card for the target sliver hank, run off a can of sliver and test its hank (Section 3b). Adjust the draft change wheel if the hank is off target.
4. **Draw frame.** Pass the sliver through breaker and finisher passages with the planned doubling (commonly six or eight slivers per delivery) and a draft approximately equal to the doubling, so hank is held while evenness improves. Retest the drawn sliver hank.
5. **Simplex.** Set the draft to convert drawn sliver hank to the planned roving hank, and roving twist just high enough for clean winding and unwinding. Test roving hank and twist.
6. **Ring frame.** Set the draft (roving hank × draft ≈ yarn count, apart from a small twist contraction) and the twist for the target count. Spin a set of cops, wind a 120-yard skein, weigh it and calculate the count produced (Section 3a).

7. **Repeat for each target count,** changing only the planned settings, and record settings and results side by side.

2.3 Procedure — jute line trial

1. **Batch and soften.** Select jute of one assortment, apply the oil-and-water emulsion evenly during softening, and leave the material to condition so the emulsion penetrates.
2. **Breaker card.** Feed the softened material to the breaker card, which breaks the long reeds down into fibre bundles and forms the first sliver. Weigh a measured length of it.
3. **Finisher card.** Pass the sliver through the finisher card for further opening and cleaning; retest sliver weight per unit length.
4. **Drawing.** Run the sliver through the successive jute drawing frames, with doubling at each passage, testing sliver weight after each.
5. **Spin.** Spin on the jute spinning frame to the target count, in pounds per spyndle (14,400 yards), and test the yarn by weighing a measured length.

2.4 Observation tables

The first row below is **sample data**, worked in Section 3; the remaining rows are for the student's own trials.

TRIAL	TARGET COUNT	LAP (G/M)	CARD SLIVER HANK	ROVING HANK	COUNT FOUND
1 (sample)	12.5 Ne	430	0.096	1.54	12.46 Ne
2	—	—	—	—	—
3	—	—	—	—	—

Record the jute trial likewise: sliver weight after each card and drawing, and the spun count in lb per spyndle.

3. Worked calculations

WORKED CALCULATION (A) — YARN COUNT FROM A WRAP-REEL SKEIN

A skein of **120 yd** wound on the wrap reel weighs **5.20 g**.

Ne = (length × 453.6) ÷ (840 × weight) = (120 × 453.6) ÷ (840 × 5.20) = 54,432 ÷ 4,368 = **12.46 Ne**, reported as **12.5 Ne** against the 12.5 Ne target.

Check: the skein is one-seventh of a full hank (840 ÷ 120 = 7), so a full hank would weigh 5.20 × 7 = 36.4 g, and 453.6 ÷ 36.4 = 12.46 — the same result.

WORKED CALCULATION (B) — SLIVER HANK FROM A MEASURED LENGTH

A length of **5 yd** of card sliver weighs **28.0 g**.

Yards per pound = (5 × 453.6) ÷ 28.0 = 2,268 ÷ 28.0 = **81 yd/lb**.

Sliver hank = 81 ÷ 840 = **0.096 hank** (0.0965 before rounding).

Note the scale: a sliver is far coarser than a yarn, so its hank is a small fraction.

WORKED CALCULATION (C) — CV% OF LAP WEIGHT, AND A ROVING HANK CHECK

Five one-metre lap specimens weigh **425, 435, 430, 445 and 415 g**.

Mean = (425 + 435 + 430 + 445 + 415) ÷ 5 = 2,150 ÷ 5 = **430 g/m**.

Deviations from the mean: −5, +5, 0, +15, −15. Squared: 25 + 25 + 0 + 225 + 225 = 500.

SD = √(500 ÷ 5) = √100 = **10.0 g**. CV% = (10.0 ÷ 430) × 100 = **2.33%**.

The mean is on target and the variation modest; a rising lap CV% at an unchanged mean is the classic early warning of a blowroom fault — uneven feed or a damaged lattice.

Roving check: 20 yd of roving weighs 7.0 g. Yards per pound = (20 × 453.6) ÷ 7.0 = 1,296 yd/lb; roving hank = 1,296 ÷ 840 = **1.54 hank**.

4. Precautions

- Condition all test material in the laboratory atmosphere before weighing; a count measured on unconditioned material measures the weather as much as the yarn.
- Let the wrap reel, not a ruler, fix every test length, and wind at even tension — a stretched skein reads long and reports a finer count than the yarn really is.
- Weigh on a checked balance; record the actual weight, never the target weight.
- Change one machine setting at a time between trials. Two changes made together cannot be separated afterwards, and the trial teaches nothing.
- On the jute line, give the emulsion its full conditioning time. Under-conditioned jute cards badly and breaks fibre, and its poor test results then blame the card for the batching department's fault.
- Keep cotton and jute material, waste and samples strictly separate; jute fibre in a cotton sliver is a serious contamination fault downstream. Guard rotating rollers, flyers and card clothing, and never reach across a running lap or sliver feed.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is the product of every spinning machine tested before it passes to the next machine, instead of testing only the finished yarn?
2. A 120-yard skein weighs 6.48 g. Calculate the yarn count in Ne.
3. Five yards of draw-frame sliver weigh 22.68 g. Calculate the sliver hank.
4. A lap has the correct mean weight per metre but its CV% has doubled since the last trial. What does that suggest, and why does the correct mean not clear the lap?
5. Why does the jute line need batching, softening and two cards, while the cotton line does not?
6. State the jute count unit, and how its sense (direct or indirect) differs from Ne.
7. Roving twist is kept low while yarn twist is high. Explain the different jobs the two twists do.

Answer key

1. Because faults are created stage by stage and are cheapest to correct where they are made. An off-hank sliver or a variable lap is partly hidden by doubling and drafting later, so a final-yarn test can pass while the process drifts, with no evidence of which stage failed.
2. Ne = (120 × 453.6) ÷ (840 × 6.48) = 54,432 ÷ 5,443.2 = **10.0 Ne**.
3. Yards per pound = (5 × 453.6) ÷ 22.68 = 2,268 ÷ 22.68 = 100 yd/lb. Hank = 100 ÷ 840 = **0.119 hank**.
4. It suggests an irregularity fault in feeding or opening — uneven feed or damaged opening parts — delivering alternately heavy and light material. Heavy and light portions cancel in the average but not in the card, which receives them in sequence and passes the variation forward into sliver unevenness.
5. Jute arrives as long, stiff, gummed reed bundles, not loose individual fibres. Emulsion softening loosens the gum and makes the fibre pliable; the breaker card breaks the reeds down and the finisher card completes the opening. Cotton arrives as comparatively clean, short staple fibre, so one carding stage suffices.
6. Jute count is in pounds per spyndle (14,400 yards): the weight of a fixed length, so a coarser yarn has the higher number — a direct system. Ne is a length per fixed weight, so a finer yarn has the higher number — an indirect system.
7. Roving twist is a handling twist — just enough to hold the roving together in winding and travel, and low enough for the ring-frame rollers to draft it out. Yarn twist is a strength twist, set far higher.